

Unit 5.4 The Fundamental Theorem of Calculus

Note Title

2012/05/12

Stewart Section 5.3 p 391-393

The mean value theorem p285

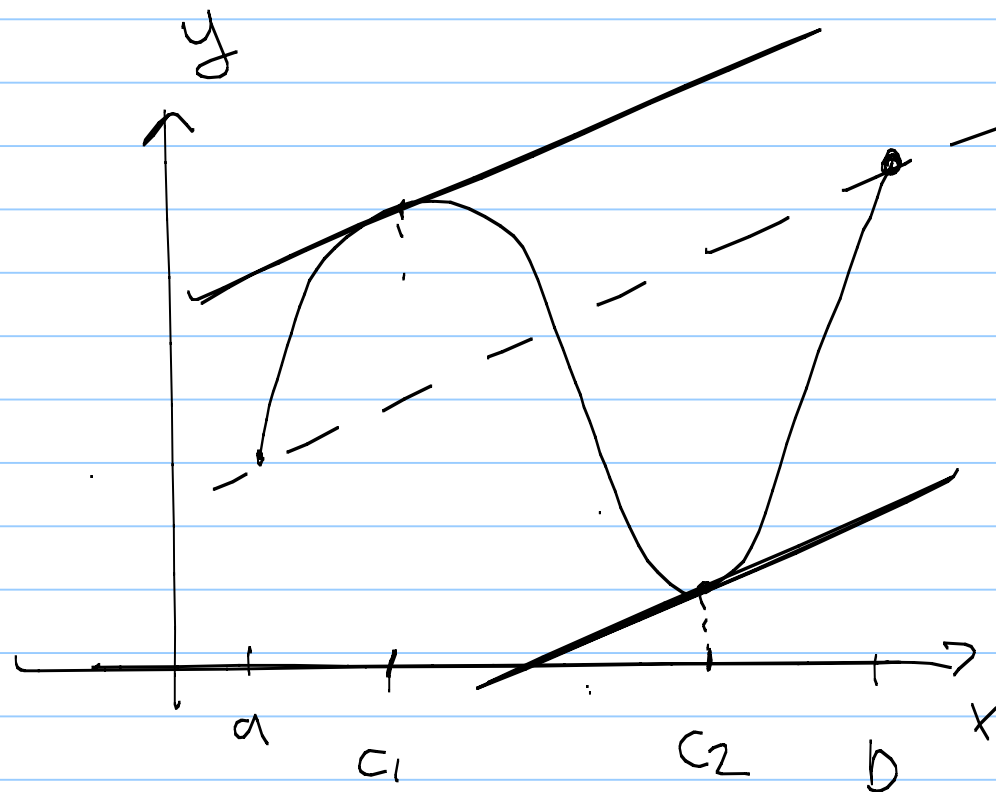
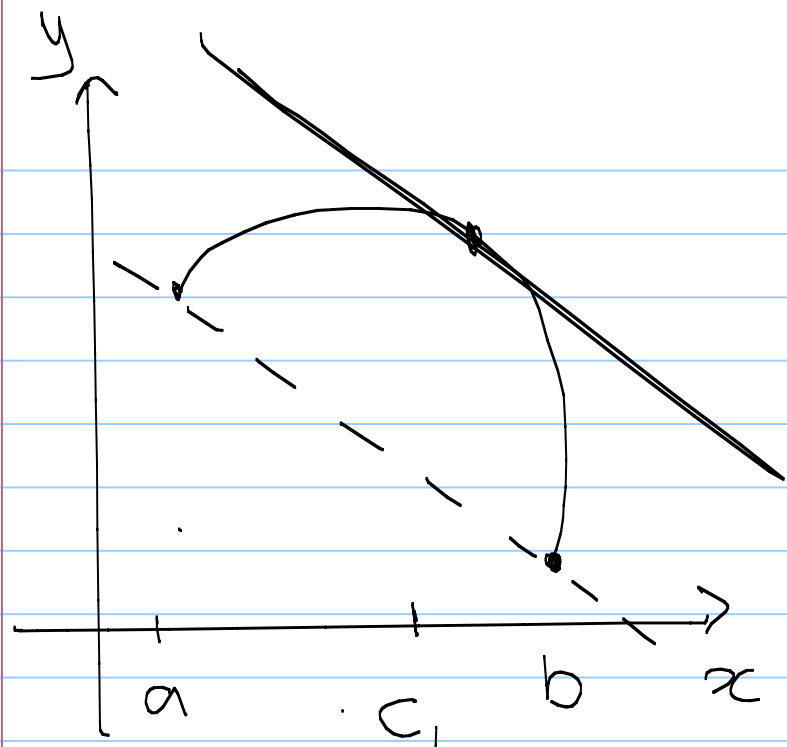
If

1. f is continuous on the closed interval $[a, b]$
2. f is differentiable on the open interval (a, b)

then there exists a $c \in (a, b)$

such that
$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

The theorem tells you that you can draw a tangent line at $x=c$ that is parallel to the secant line through the point $(a, f(a))$ and $(b, f(b))$



The Fundamental Theorem of Calculus

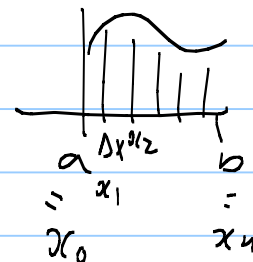
If f is continuous on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f , that is $F' = f$.

Proof

Divide $[a, b]$ into n subintervals of equal length $\Delta x = \frac{b-a}{n}$.



Let $x_0 = a < x_1 < \dots < x_{n-1} < x_n = b$

be the endpoints of these intervals — (A)

Let F be an antiderivative of f ,

that is F is a function such that $F' = f$ — (B)

That means

- F is differentiable on $[a, b]$
(as its derivative is equal to f , the derivative must exist)

- F is continuous on $[a, b]$

By Thm 4 pg 188

If f is differentiable at a ,
then f is continuous at a

Use the Mean Value Theorem
on each of the subintervals
 $[x_{i-1}, x_i]$: Since

1. f is continuous on $[x_{i-1}, x_i]$

2. f is differentiable on
 (x_{i-1}, x_i)

There exists a number c_i in
the interval (x_{i-1}, x_i) such
that

$$\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(c_i) \rightarrow \textcircled{C}$$

from $\textcircled{B}$

Let these c_i be the sample point in each interval.

As f is continuous on $[a, b]$,
 f is integrable on $[a, b]$

Thm p 373

If f is continuous on $[a, b]$, then
 f is integrable on $[a, b]$

Using definition on p 372

$$\int_a^b f(x) dx$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} \Delta x$$

from (C)

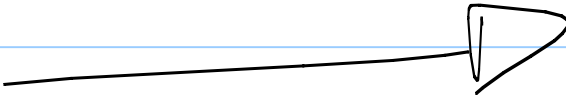
$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} (x_i - x_{i-1})$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n F(x_i) - F(x_{i-1})$$

$$= \lim_{n \rightarrow \infty} \left([F(x_1) - F(x_0)] + [F(x_2) - F(x_1)] \right. \\ \left. + \dots + [F(x_n) - F(x_{n-1})] \right)$$

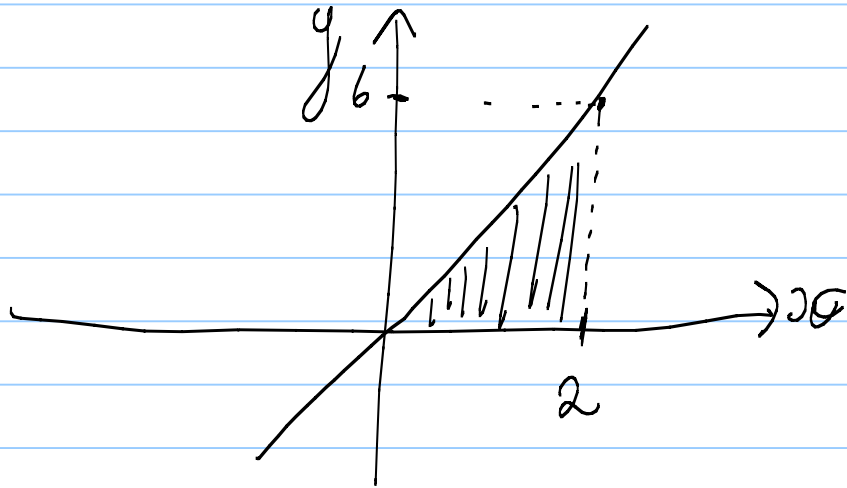
$$= \lim_{n \rightarrow \infty} [F(x_n) - F(x_0)]$$

$$= \lim_{n \rightarrow \infty} \left([F(b) - F(a)] \right) \quad \begin{matrix} \text{From} \\ (A) \end{matrix}$$

$$= F(b) - F(a)$$


Example. Calculate the area

$$1. \int_0^2 3x \, dx$$

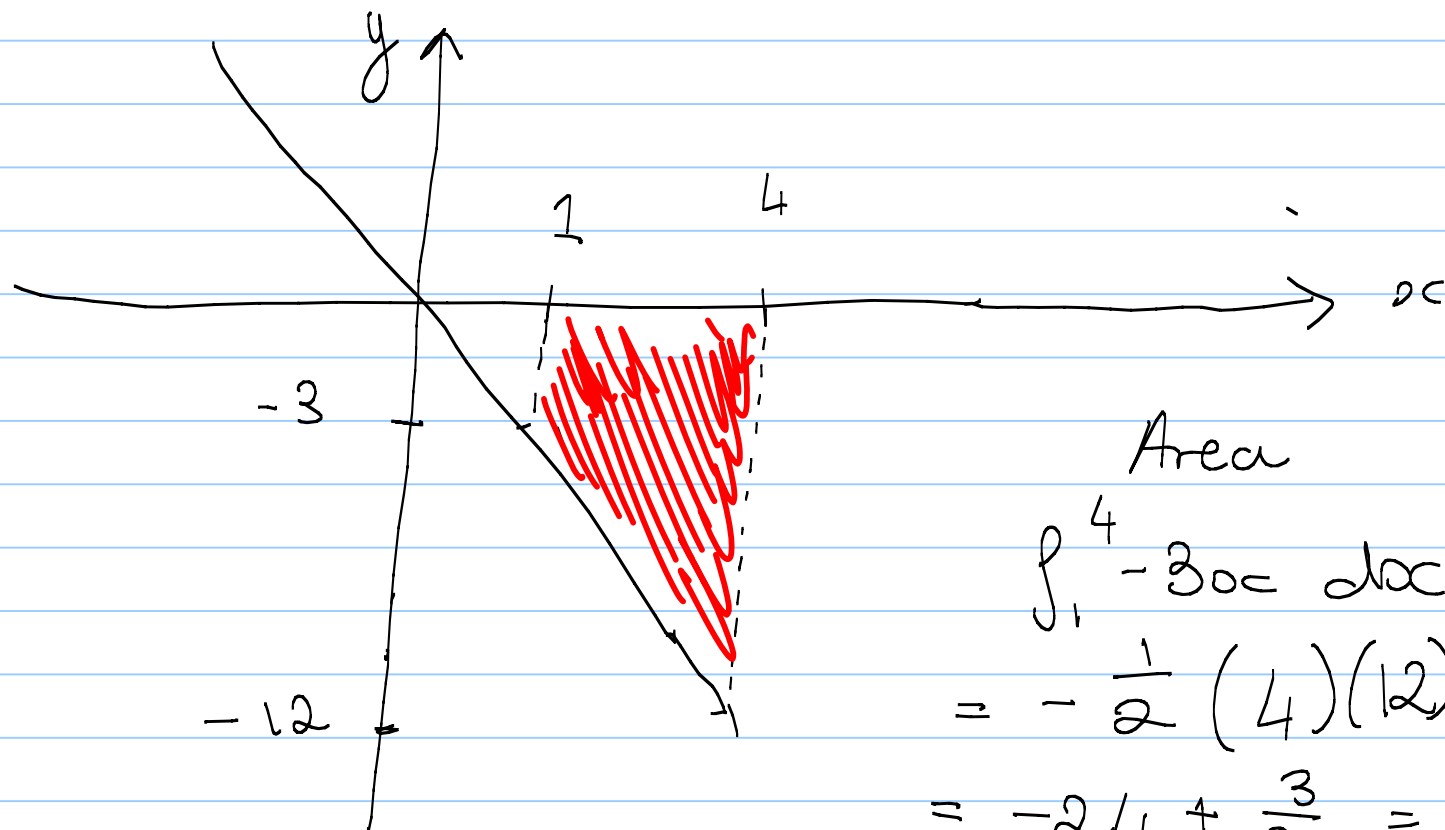


$$\int_0^2 3x \, dx = \frac{1}{2} (2)(6) = 6$$

$$\begin{aligned}
 \int_0^2 3x \, dx &= \frac{3}{2} x^2 \Big|_0^2 \\
 &= \frac{3}{2} (2^2 - 0^2) \\
 &= \frac{3}{2} \cdot 4 \\
 &= 3 \cdot 2 = 6
 \end{aligned}$$

The Same!

Q. $\int_1^4 -3x \, dx$



Area

$$\begin{aligned}
 & \int_1^4 -3x \, dx \\
 &= -\frac{1}{2} (4)(12) + \frac{1}{2} (1)(3) \\
 &= -24 + \frac{3}{2} = -\frac{45}{2}
 \end{aligned}$$

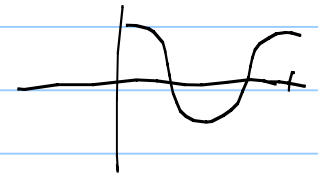
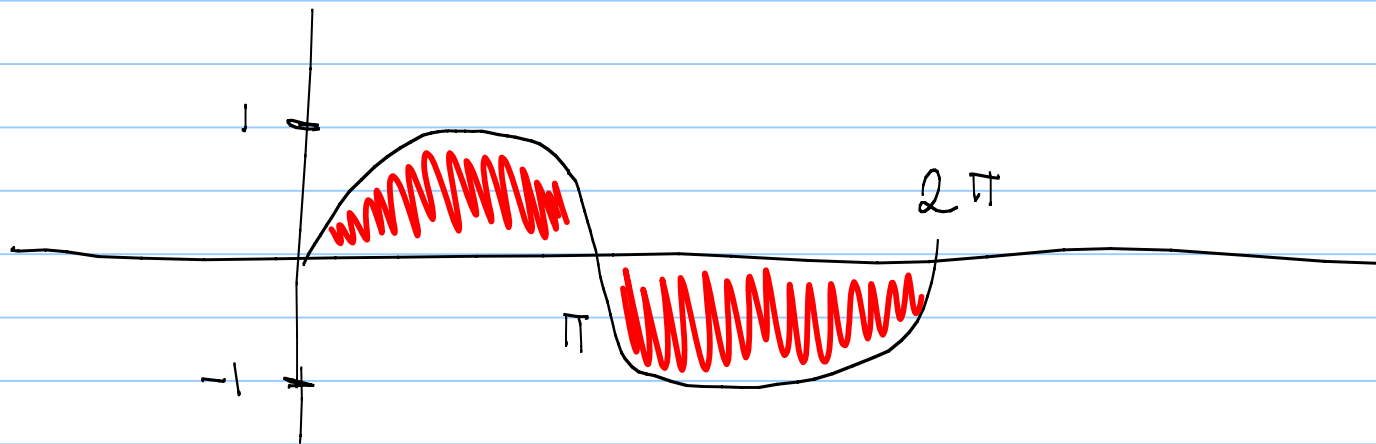
$$\int_1^4 -3x \, dx$$

$$= -\frac{3}{2} x^2 \Big|_1^4$$

$$= -\frac{3}{2} (4^2 - 1^2)$$

$$= -\frac{3}{2} (16 - 1) = -\frac{3}{2} (15) = -\frac{45}{2}$$

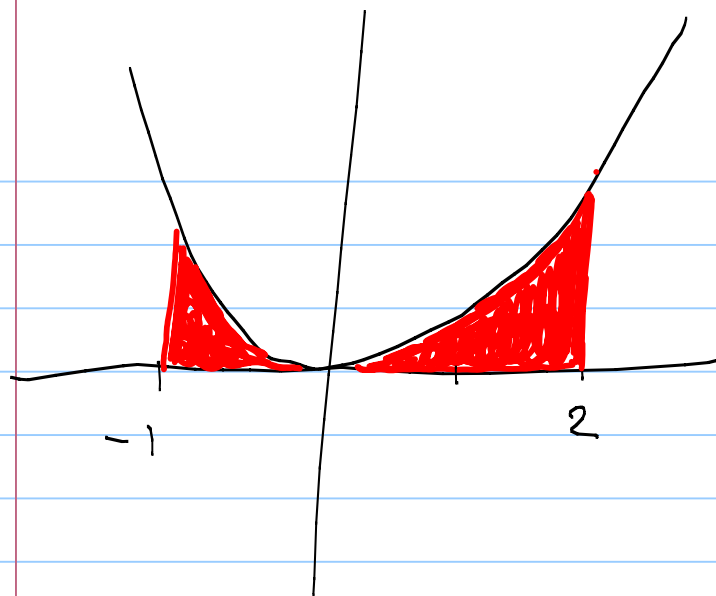
$$\int_0^{2\pi} \sin x \, dx = 0$$



$$\begin{aligned} \int_0^{2\pi} \sin x \, dx &= -\cos x \Big|_0^{2\pi} \\ &= -[\cos 2\pi - \cos 0] = -[1 - 1] = 0 \end{aligned}$$

Example

1] Calculate the area of the region between the graph of $f(x) = x^2$ and the x -axis between $x = -1$ and $x = 2$.

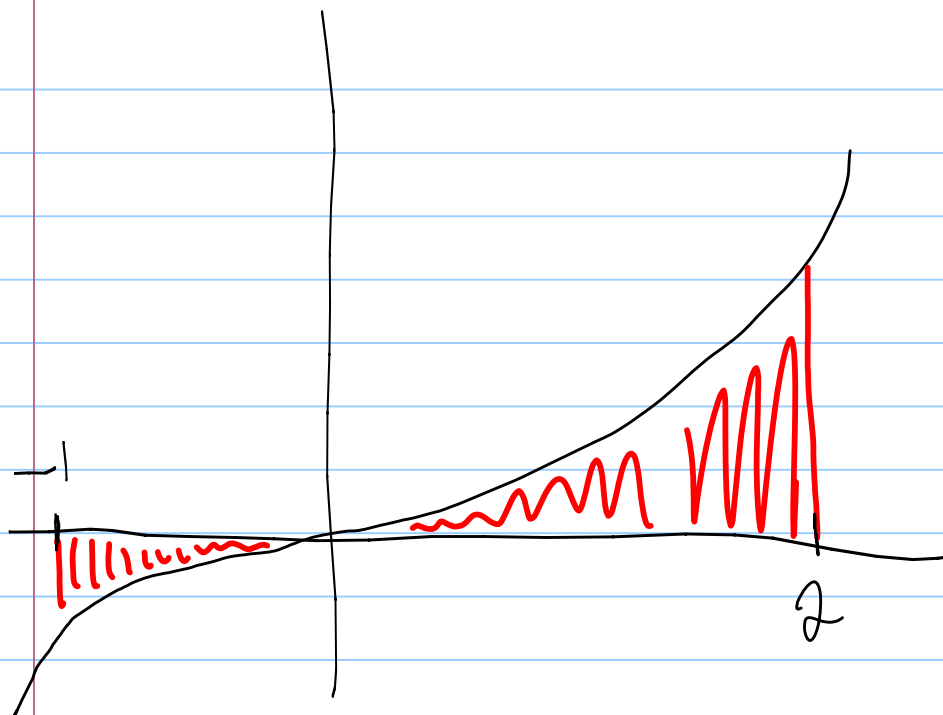


$$\int_{-1}^2 x^2 dx$$
$$= \frac{1}{3} x^3 \Big|_{-1}^2$$

$$= \frac{1}{3} (2)^3 - \frac{1}{3} (-1)^3$$

$$= \frac{1}{3} 8 + \frac{1}{3} = 3$$

2] Calculate the area of the region between the graph of $f(x) = x^3$ and the x -axis between $x = -1$ and $x = 2$.



$$\int_{-1}^0 x^3 dx$$

$$= \frac{1}{4} x^4 \Big|_{-1}^0$$

$$= 0 - \frac{1}{4} = -\frac{1}{4}$$

$$\int_0^2 x^3 dx$$

$$= \frac{1}{4} x^4 \Big|_0^2$$

$$= \frac{16}{4} - 0 = 4$$

$$\text{Area} = \frac{1}{4} + 4 = \frac{17}{4}$$

Exemples

Calculate

$$1. \int_{-1}^2 e^{5x} dx$$

$$2. \int_{-2}^1 x^2 e^{x^3} dx$$

$$3. \int_0^{\pi} \cos 3x dx$$

$$4. \int_0^1 2^t \cos 2^t dt$$

$$5. \int_{-1}^1 x \sqrt{1+x^2} dx$$

$$6. \int_{-2}^{-1} \sqrt{1-x} \, dx$$

Solutions

$$1. \int_{-1}^2 e^{5x} dx$$

$$= \left. \frac{1}{5} e^{5x} \right|_{-1}^2$$

$$= \frac{1}{5} e^{10} - \frac{1}{5} e^{-5}$$

$$2. \int_{-2}^1 x^2 e^{x^3} dx$$

$$= \frac{1}{3} e^{x^3} \Big|_{-2}^1$$

$$= \frac{1}{3} e - \frac{1}{3} e^{-8}$$

$$3. \int_0^{\pi} \cos 3x \, dx$$

$$= \frac{1}{3} \sin 3x \Big|_0^{\pi}$$

$$= \frac{1}{3} \sin 3(\pi) - \frac{1}{3} \sin 3(0)$$

$$= 0$$

$$4. \int_0^1 2^t \cos 2^t dt$$

$$= \sin 2^t \cdot \frac{1}{\ln 2} \Big|_0^1$$

$$= \frac{\sin 2}{\ln 2} - \frac{\sin 1}{\ln 2}$$

$$5. \int_{-1}^1 x \sqrt{1+x^2} \, dx$$

$$= \frac{2}{3} (1+x^2)^{\frac{3}{2}} \cdot \frac{1}{2} \Big|_{-1}^1$$

$$= 0$$

$$6. \int_{-2}^{-1} \sqrt{1-x} \, dx$$

$$= -\frac{2}{3} (1-x)^{3/2} \Big|_{-2}^{-1}$$

$$= -\frac{2}{3} (2)^{3/2} - \left(-\frac{2}{3} (3)^{3/2} \right)$$

$$= -\frac{2}{3} \sqrt{2^3} + \frac{2}{3} \sqrt{3^3}$$

$$= -\frac{2}{3} \sqrt{8} + \frac{2}{3} \sqrt{27}$$

$$= -\frac{4}{3} \sqrt{2} + 2 \sqrt{3}$$
